TECHNICAL **CATALOG**

Technology & Products

Innovation,
Performance & Safety

cvdentus

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THE COMPANY

Research on synthetic diamond began in 1991, when scientist Vladimir Jesus Trava Airoldi resumed his work as a researcher at the National Institute for Space Research (Inpe) after completing his postdoctoral studies at NASA, in Pasadena, California.

Back in Brazil, in 1997 he founded the company CVDVale together with a group of scientists, whose goal was to bring to society the benefits of years of research and cutting-edge technology from Space Research. Dentistry was chosen as the first industrial application of artificial diamond because dentists are professionals who tirelessly seek a technological edge for their practice and patients.

Today, beyond the dental market, the company also operates and adds value in other segments:

cvdvale

cvdentus	cvdinstruments	cvdcoating	cvdrill
Dental	Scientific and Industrial	Industrial	Mining, Oil & Gas

To this day, CVDentus is **the only brand** in the world to use CVD diamond in Dentistry.

It also has:

4 PhDs	*+ 100 Scientific Articles*	*+ 15 Patents*
+ 30 representatives	*+ 5 countries*	*+ 10 Awards*

TIMELINE

1989

1991

Founded the company CVDVALE in **São José dos Campos - SP** through funding from FAPESP's Technological Innovation Program

1997

The CVD Diamond dental bur patent won the State Governor's Award for invention of the year.

1998

2003

FINEP Technological Innovation Award and honorable mention presented personally by the President.

Launch of the **CVDent1000**, CVDentus' first dental ultrasound.

2007

2011

World's best invention award, for the first time in the Medical-Dental field **(WIPO)**

Launch of the new clinical ultrasound equipment, DentSurg.



2012

2014

Launch of the DentSurg Surgical, the first "2 in 1" piezoelectric ultrasonic unit with clinical and surgical functions.

Launch of the DentSurg Pro Piezo and of surgical tips made of aeronautical-grade titanium alloy.



2018



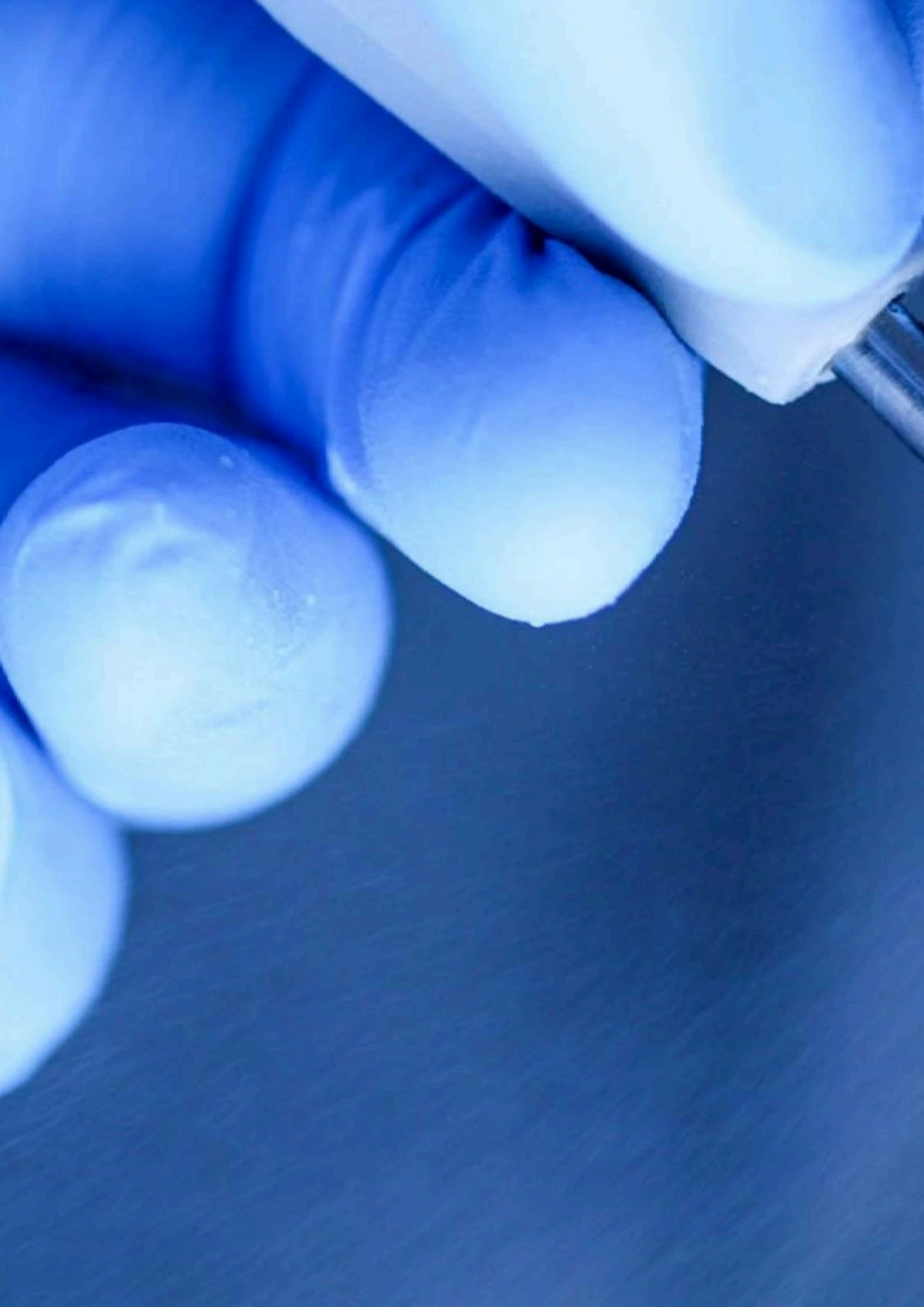
Launch of the OPTYMUS, the smallest and most complete Piezo in its category.

2019



Launch of the Clinical Plus Piezo.

2020



TECHNOLOGIES



Foto: Dr. Rafael Moraes

TECHNOLOGIES

CVD diamond is synthetic and has the same physical and chemical properties as natural diamond: extreme hardness, excellent thermal conductivity, very low friction coefficient (equivalent to Teflon), biological compatibility, chemical inertia at temperatures below 800°C, among others.

The CVD diamond production technique is chemical vapor deposition assisted by hot filaments, also known as HFCVD (hot-filament Chemical Vapor Deposition), and in this system a single, homogeneous stone of predetermined thickness and grain size grows directly on the active part of the tip. In other words, the integrity of the diamond and its high adhesion to the metal shank are guaranteed by this method, thereby adding to the part all the advantages of diamond already mentioned here. This internationally patented technology is 100% sustainable and well known for significantly increasing the service life of tips compared to traditional ones.

The differentiators, however, are not limited to physical and chemical aspects. Beyond durability, CVD diamond technology used for dental purposes, a CVDentus exclusive, also favors the practice and clinical routine of dentists.

Conventional Diamond

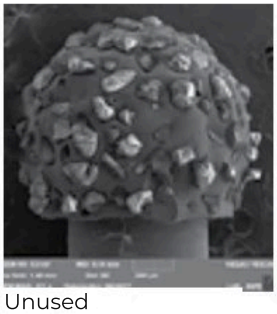


CVDentus Diamond

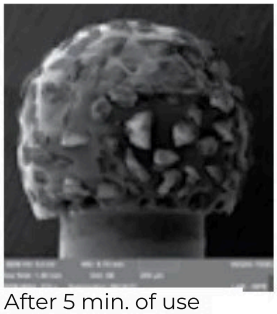


CVD DIAMOND - CHEMICAL VAPOR DEPOSITION

Conventional diamond-coated tip



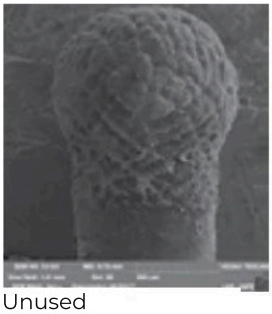
Unused



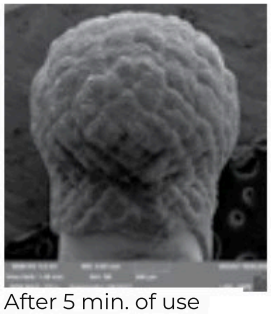
After 5 min. of use

- Diamond powder
- Durability 30x lower than CVD diamond
- Irregular surface with sharp edges
- Irregular finish
- Generates Smear Layer
- Direct contact with metal

CVDentus diamond tip



Unused



After 5 min. of use

- Single diamond stone
- High durability
- Homogeneous surface, free of edges
- Superior, homogeneous finish
- Generates less Smear Layer
- Contact only with diamond, biologically compatible

ULTRA FINE | Ideal for CAD/CAM and lenses, Perio and pre-polishing preparation.

FINE | Medium roughness, used for removal of resin and general finishing.

COARSE | Cavity preparations, finish lines for prostheses and lenses, and general wear.



TECHNOLOGIES



Photo: Prof. Paulo Kano

DOES NOT CUT SOFT TISSUE

CVD diamond has a smooth, homogeneous surface that, combined with the vibration movement of ultrasound, preserves gingiva, nerves, vessels, periosteum, tongue and cheeks.



Photo: Prof. Ricardo Bertholdi

CUTS ENAMEL, DENTIN, AMALGAM, RESIN AND BONE

CVD diamond has a hardness of 95GPa, that is, 30x higher than tooth enamel, and therefore cuts extremely rigid structures.



Photo: Prof. Renato Voss

EFFICIENCY

The cutting power of the tips remains the same until the end of their life cycle, unlike burs, which lose cutting ability as the grains come loose. They also provide predictability and agility in procedures, ensuring that several treatments are completed in fewer steps or with a significant reduction in recovery interval time.

DIFFERENTIALS & BENEFITS OF CVD DIAMOND

PRECISION

The design of the CVD diamond tip, together with the ultrasonic effect, enables minimally invasive treatments, clean surgeries with greater predictability, giving the patient more comfort and a faster recovery.



Photo: Prof. Gustavo Passarelli Petris

PAIN-FREE IN UP TO 70% OF TREATMENTS

A study compared 34 class V cavity preparations, performed 50% with the CVDentus system and 50% with high-speed burs. The results showed that in those performed with the CVDentus system there was no damage to the pulp-dentin membrane or to the layers of odontoblasts, dentin, predentin and cell-rich zone, which were preserved; in the cases performed with diamond burs and high speed, all showed lack of the pulp-dentin membrane, disarrangement of the intratubular fluid and stimulation of nerve endings.



Photo: Prof. Renato Voss

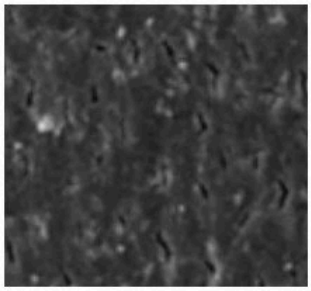
BIOCOMPATIBILITY

The patient has no contact with toxic substances such as Nickel, Silicon, Chromium and Iron, found in conventional tips.

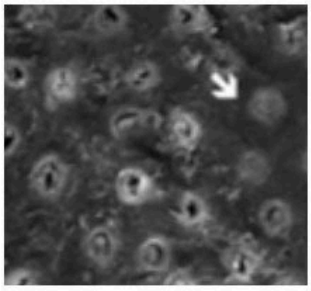


Photo: Prof. Thiago Ottoboni

TECHNOLOGIES



High Speed



CVDentus Tip

GENERATES LESS SMEAR LAYER

Studies indicate that CVDentus tips generate a less adherent smear layer and therefore show superior dentin conditioning compared to conventional preparations.



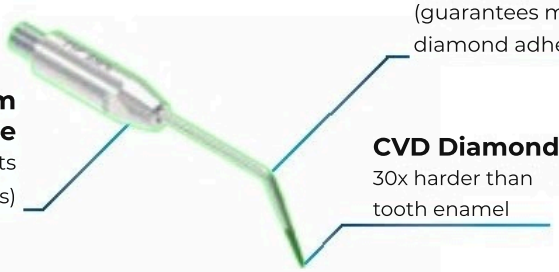
Photo: Prof. Renato Voss

BETTER FINISH

Finally, together with ultrasound, CVD diamond technology also ensures greater visibility, has no high-speed noise, offers greater biosafety to the professional and also speeds up the postoperative period.

CVDENTUS TIPS

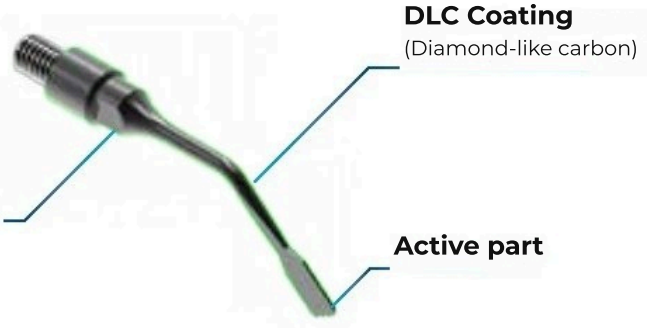
CLINICAL TIPS WITH CVD DIAMOND



CLINICAL TITANIUM TIPS



CVDENTUS SURGICAL TIPS



TECHNOLOGIES

Diamond-Like Carbon (DLC) is a thin carbon film with physical-chemical characteristics close to those of diamond. The application of this film is broad, since it can be deposited on metallic and non-metallic surfaces of large areas and complex geometries.

At CVDentus, it is used to coat the surgical tips.





Photos: Prof. Diego Bazan

DLC DIAMOND - DIAMOND LIKE CARBON

The characteristics inherent to DLC are:

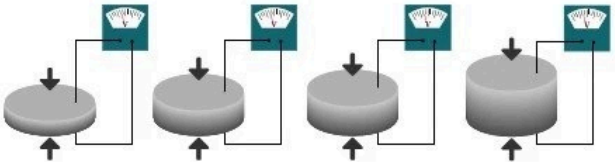
- Durability and resistance
- Bactericidal effect
- Low friction coefficient
- Lower likelihood of bone necrosis
- Chemically inert material
- Procedures free of metal residues
- Easy to sterilize
- Anti-coagulant



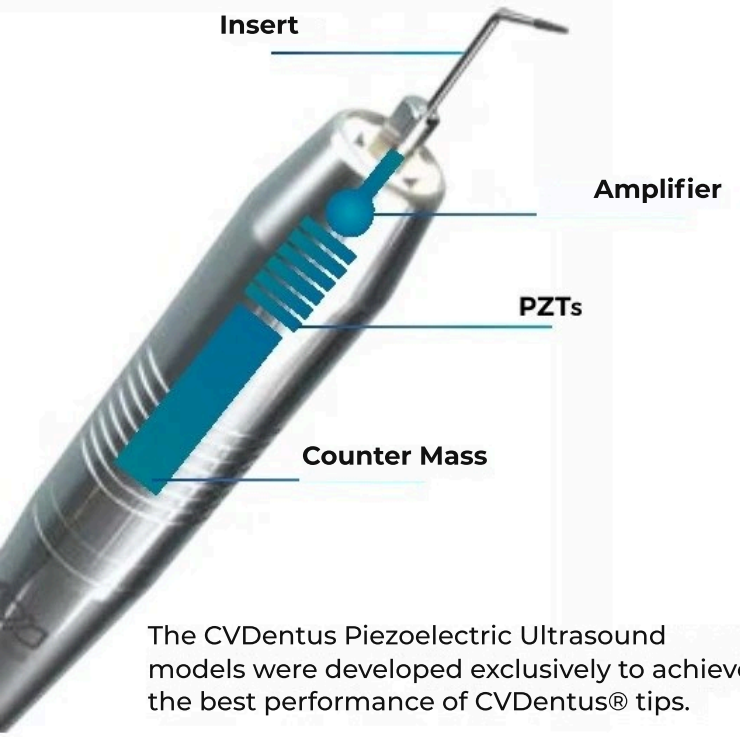


PIEZO

Piezo is the name of the crystal used to translate electrical pulses into mechanical vibration. The piezoelectric effect is the ability of some crystals to generate electrical voltage in response to mechanical pressure and vice versa. In dental equipment, Piezos are those that have piezoelectric transducers in their system, such as PZT-type ceramic pellets. When subjected to electricity, this material undergoes volumetric changes and converts electrical impulses into ultrasonic vibration.



Conversion of electrical pulses into mechanical movement in a piezoelectric crystal.





The CVDentus Piezoelectric Ultrasound models were developed exclusively to achieve the best performance of CVDentus® tips.

THE COMPLETE SOLUTION IN PIEZOS

DentSurg PRO

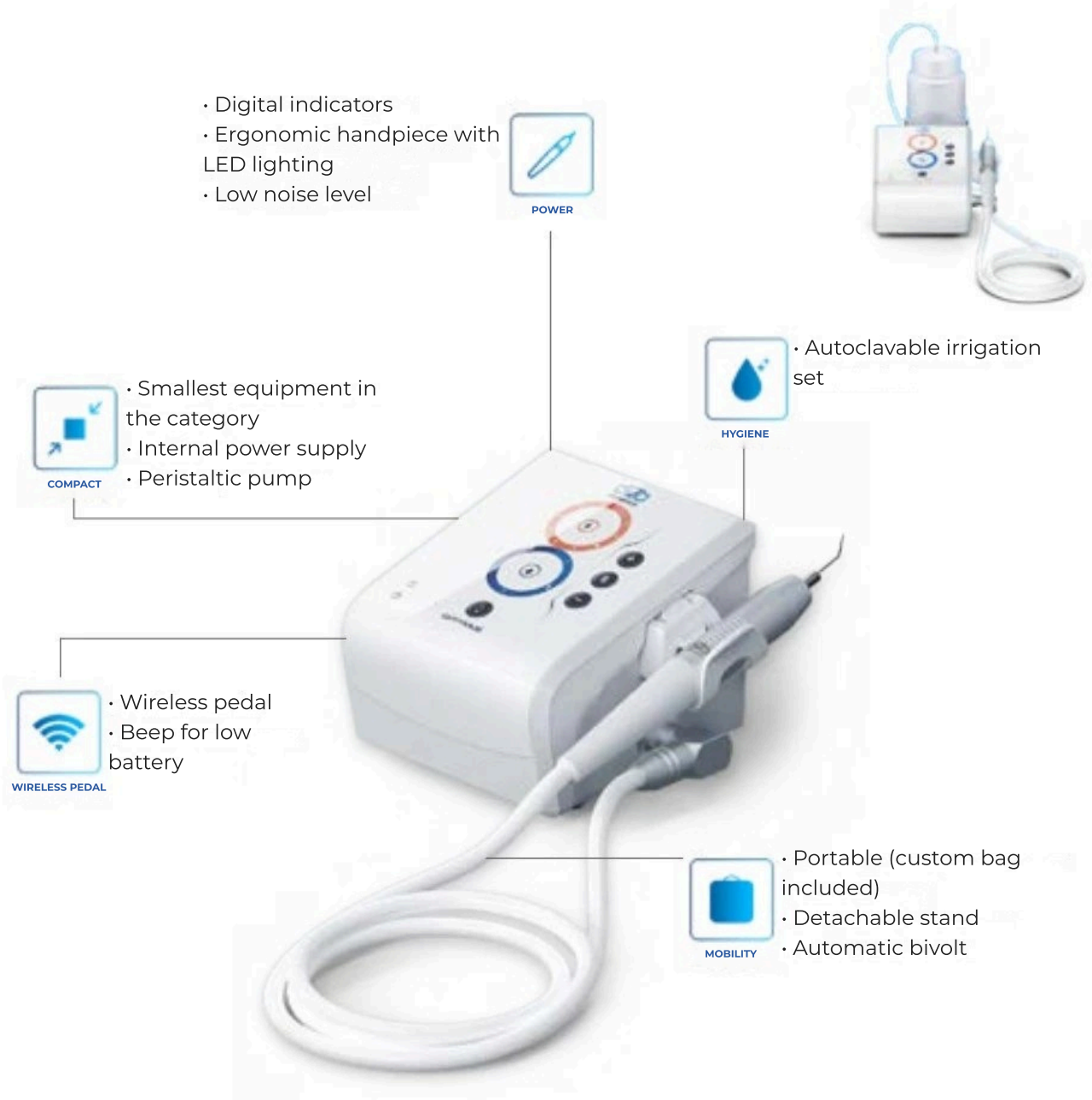
Clinical PLUS



OPTYMUS

The smallest and most complete in its category.

The only Piezo with exclusive differentiators in its category. Compact, portable and easy to handle, it benefits the daily routine of professionals who value quality in procedures and comfort for their patients.



· This equipment has protection against power grid fluctuations, Thermostat and Frequency Stabilizer.

ENDO | ENDODONTICS

PERIO | PROPHYLAXIS

Clinical PLUS

Essential. In every sense.

The complete Piezo for clinical specialists. This technology gives treatments predictability, comfort to patients, and agile, safe and minimally invasive procedures.



- 5" touchscreen
- Intuitive menu
- Automatic and manual programming

- Ergonomic handpiece with adjustable 3-level LED lighting
- Autoclavable irrigation set

- Software update via USB

WIRELESS PEDAL

- Detachable pedal cable
- Beep for low battery

- Portable (carrying bag option)
- Internal power supply
- Detachable stands
- Automatic bivolt
- Low noise level

- This equipment has protection against power grid fluctuations, Thermostat and Frequency Stabilizer.

ENDODONTICS | PERIODONTICS | RESTORATIVE | PROSTHODONTICS

DentSurg PRO

Precision with high performance.
The evolution of the Piezo.

The only Piezo that serves all specialties. Designed to achieve maximum performance, it also allows exclusive clinical procedures and surgeries with fast and precise bone cuts, both without damaging soft tissues. There is no shortage of arguments for becoming a PRO.



- 7" touchscreen
- Intuitive menu
- Automatic and manual programming
- Software update via USB

- Ergonomic handpiece with adjustab
level LED lighting
- Autoclavable irrigation set

- High cutting performance

- Detachable stands
- Automatic bivolt
- Low noise level

- Portable (carrying bag option)
- Internal power supply

- This equipment has protection against power grid fluctuations, Thermostat and Frequency Stabilizer.

ENDO ENDODONTICS

PERIO PERIODONTICS

CAVITY RESTORATIVE PROSTHODONTICS

SURGERY CORTICAL MEDULLARY MEMBRANE

TIP MODELS

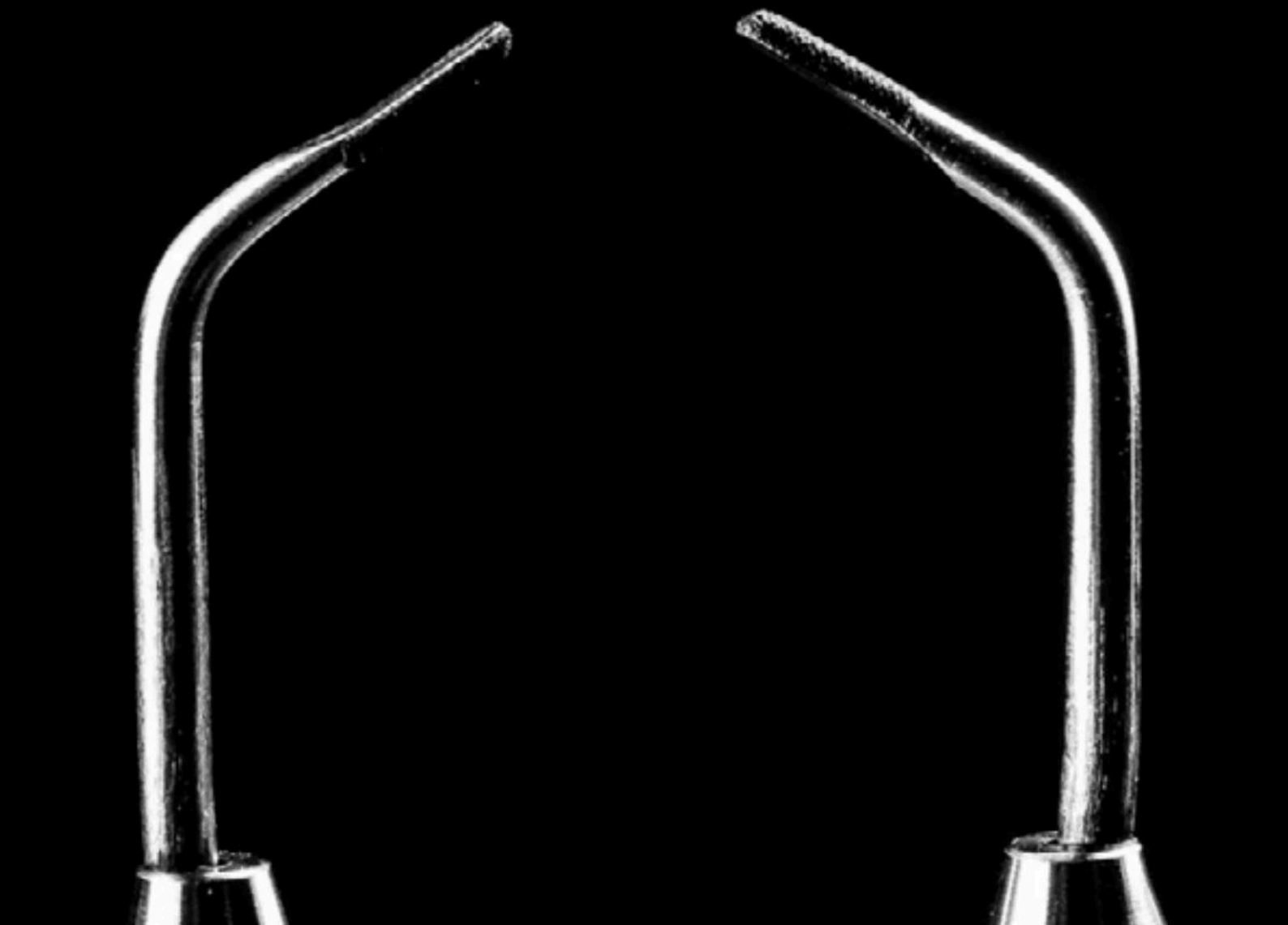


Foto: Prof. Ricardo Bertholdi

OPTYMUS



ENDO



PERIO

PROPHYLAXIS





C22-E

Ø 1.2mm

- Pulp chamber preparation
- Angle preparation



C2T

Ø 1.6mm

- Post removal



CR4F-ST

Ø 1.6mm

- Smoothing of pulp chamber walls without wearing the floor



R1-E

Ø 1.0mm

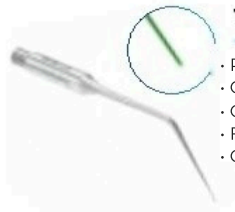
- Pulp chamber access / preparation
- Removal of pulp stones in the pulp chamber
- Compacting and flattening gutta-percha



T0T

Ø 0.3mm

- Conventional supra- and subgingival scaling



T0T-E1

Ø 0.3mm

- Passive irrigation (PUI)
- Cement activation
- Cutting and thermoplasticization of gutta-percha
- Fractured file removal
- Cleaning of isthmus areas



T0T-E2

Ø 0.3mm

- Passive irrigation (PUI)
- Cement activation
- Cutting and thermoplasticization of gutta-percha
- Fractured file removal
- Cleaning of isthmus areas



T0T-E3

Ø 0.45mm

- Cleaning of flattened canals
- Cleaning of canals in retreatment



T1F-2

Ø 0.5mm

- Root-end preparation surgery in posterior teeth



T1F-E

Ø 0.5mm

- Canal location and clearing
- Pulp stone/isthmus removal



T1F-L

Ø 0.5mm

- Root-end preparation surgery in posterior teeth



T1F-R

Ø 0.5mm

- Root-end preparation surgery in posterior teeth



T1T

Ø 0.6mm

- Conventional supragingival scaling



T1U-2

Ø 0.5mm

- Scaling in the furcation area



TR1F-ST

Ø 0.5mm

- Wear in the isthmus region and refinement of the canal entrance



V1-E

Ø 1.2mm

- Flattening of pulp chamber walls



W1

0.5mm x 2.5mm

- Bone window opening
- Apicoectomy



W1T-O

0.4mm x 2.5mm

- Conventional supragingival scaling



W1-0

0.5mm x 2.5mm

- Bone window opening
- Apicoectomy





Photos - Prof. Rodrigo Villagra: Pulp chamber cleaning and cutting and thermoplasticization of gutta-percha



"Together with CVDentus tips, ultrasound is a tool of great importance for those seeking excellence in endodontic treatment, as it helps in several steps, such as: conservative access surgery, cavity cleaning, PUI, post removal, retreatment, among others.

PROF. **RODRIGO VILLAGRA**

Endodontics specialist

Clinical PLUS



DentSurg PRO



PROSTHODONTICS
RESTORATIVE





RESTORATIVE

Clinical PLUS



CAVITY

DentSurg PRO



C1

Ø 1.2mm

- Angled cavity preparation
- Beveling



C22-E

Ø 1.2mm

- Angle preparation



CH1-D

1.0mm x 0.5mm

- Small cavity preparation in the proximal region



CH1-M

1.0mm x 0.5mm

- Cavity preparation in the proximal region (mesial)



CH4-D

1.6mm x 0.8mm

- Cavity preparation in the proximal region (distal)



CH4-M

1.6mm x 0.8mm

- Cavity preparation in the proximal region (mesial)



CR4

Ø 1.6mm

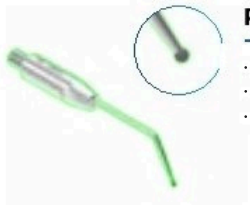
- Removal of post-bracket resin (thick layer)



R0,5

Ø 0.5mm

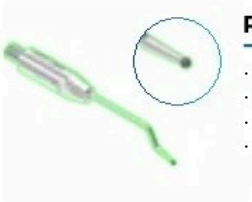
- Removal of small caries



R1

Ø 1.0mm

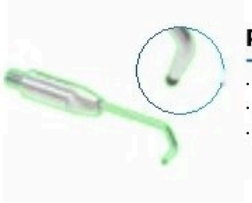
- Caries removal
- Cavity preparation in the proximal region
- Papillary contouring



R1-A

Ø 1.0mm

- Caries removal
- Cavity preparation in the proximal region
- Cavity preparation in the buccal region
- Tunnel cavity preparation with mesial access



R1-B

Ø 1.0mm

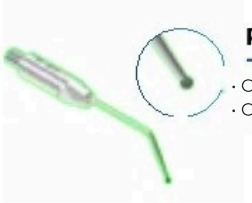
- Caries removal
- Tunnel cavity preparation with full access
- Cavity preparations on the lingual surface of anterior teeth



R1-E

Ø 1.0mm

- Tunnel cavity preparation
- Papillary contouring



R2

Ø 1.2mm

- Caries removal
- Cavity preparation



R4

Ø 1.5mm

- Removal of large caries
- Cavity preparation



“In Pediatric Dentistry I recommend the use of the CVDentus R0.5 diamond tip: developed with high technology, it has the perfect size and angulation for small cavity preparations, enabling minimally invasive procedures.

invasivos.”

DR. **LIETE YASSUDA**

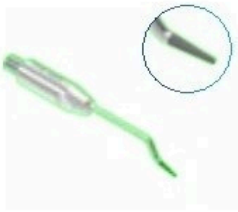
Specialist in Pediatric Dentistry



T1

Ø 0.5mm

- General cavity preparation
- Removal of amalgam or restorative resin
- Groove preparation with more incipient lesions
- Removal of excess restorative material



T1A

Ø 0.5mm

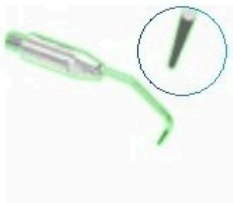
- Root sectioning



T1F

Ø 0.5mm

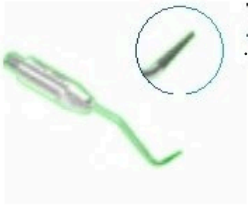
- Resin finishing
- Completion of post-bracket resin removal
- Removal of excess restorative material in the interpapillary region, supra- and subgingival



T1F-2

Ø 0.5mm

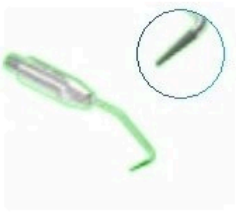
- Removal of excess restorative material in the interpapillary region, supra- and subgingival



T1F-L

Ø 0.5mm

- Removal of excess restorative material in the interpapillary region, supra- and subgingival



T1F-R

Ø 0.5mm

- Removal of excess restorative material in the interpapillary region, supra- and subgingival



W1

0.5mm x 2.5mm

- Removal of amalgams and resins (large)
- Slice



W1-O

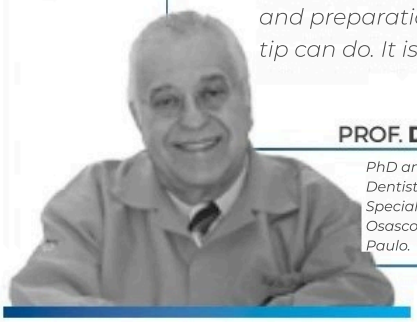
0.5mm x 2.5mm

- Removal of amalgams and resins (large)





Photos: Prof. Gustavo Petris - Molar wear in a hard-to-reach area while preserving the adjacent tooth



"With CVDentus diamond, it is possible to perform treatments and preparations that no other tip can do. It is spectacular!"

PROF. DIRCEU VIEIRA

PhD and Master's in Restorative Dentistry, Coordinator of the Specialization courses at ABO Osasco, Santos and ABS - São Paulo.

"Safety and precision are all I need in my daily practice as a general practitioner. Being able to work with CVDentus makes every procedure I perform easier, from the simplest to the most complex, in addition to being completely portable and having an innovative and bold design."

PROF. GUSTAVO PETRIS

Master's in Clinical Dentistry - Biomaterials and Specialist in Implantology.





ENDO

Clinical PLUS

DentSurg PRO



C22-E

Ø 1.2mm

- Pulp chamber preparation



C2T

Ø 1.6mm

- Post removal



CR4F-ST

Ø 1.6mm

- Smoothing of pulp chamber walls without wearing the floor



R1-E

Ø 1.0mm

- Pulp chamber access and preparation
- Removal of pulp stones in the pulp chamber
- Compacting and flattening gutta-percha



R2

Ø 1.2mm

- Pulp chamber access



R4

Ø 1.5mm

- Pulp chamber access



T0T-E1

Ø 0.3mm

- Passive irrigation (PUI)
- Cutting and thermoplasticization of gutta-percha
- Fractured file removal



T0T-E2

Ø 0.3mm

- Passive irrigation (PUI)
- Cutting and thermoplasticization of gutta-percha
- Fractured file removal



T0T-E3

Ø 0.45mm

- Cleaning of flattened canals
- Cleaning of canals in retreatment



T1F-2

Ø 0.5mm

- Root-end preparation surgery in posterior teeth



T1F-E

Ø 0.5mm

- Canal location and clearing
- Pulp stone/isthmus removal



T1F-R

Ø 0.5mm

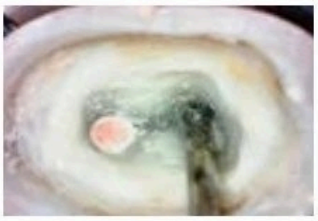
- Root-end preparation surgery in posterior teeth



“These 2 inserts are very interesting: the TR1F-ST works in the isthmus region and refines the canal entrance, and the CR4F-ST finishes and smooths the pulp chamber walls, ensuring the safety of the procedures and extreme quality in the preparations.

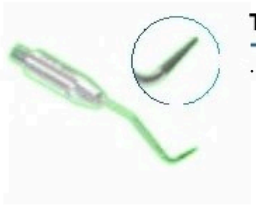
PROF. **RODRIGO VILLAGRA**

Endodontics specialist.





Photos - Prof. Rodrigo Villagra: Cutting and thermoplasticization of Gutta-Percha



T1F-L

Ø 0.5mm

· Root-end preparation surgery in posterior teeth



TR1F-ST

Ø 0.5mm

· Wear in the isthmus region and refinement of the canal entrance



V1-E

Ø 1.2mm

· Flattening of pulp chamber walls



W1

0.5mm x 2.5mm

· Apicoectomy



W1-O

0.5mm x 2.5mm

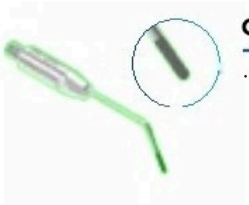
· Tooth sectioning



PERIO

Clinical PLUS

DentSurg PRO



CR1 ∅ 1.2mm

· Gingival peeling



CR4 ∅ 1.6mm

· Gingival peeling



FS-O 0.4mm x 2.5mm

· Crown lengthening with flap opening (bone volume reduction)



T0T ∅ 0.3mm

· Conventional supra- and subgingival scaling



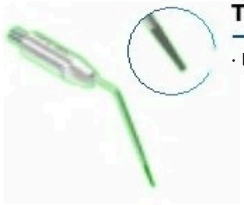
T1T ∅ 0.6mm

· Conventional supragingival scaling



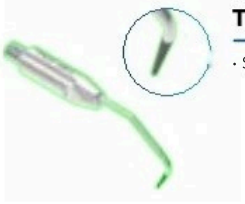
T1U ∅ 0.5mm

· Supragingival scaling with diamond



T1F-E ∅ 0.5mm

· Root planing



T1U-2 ∅ 0.5mm

· Scaling in the furcation area



T1U-E ∅ 0.5mm

· Subgingival scaling with diamond





Photos: Prof. Diego Bazan: Gingival peeling and postoperative



"I use diamond tips because they are better, effectively proven. Better than laser peeling surgery or with an Orban scalpel. The pigments take a long time to reappear and, in many cases, never appear again.

DR. **DÉBORA AYALA**

Master's in Laser, Specialist in Periodontics, Specialist in Restorative Dentistry and Specialist in Implantology.



"Among its many applications, CVDentus tips also allow melanoplasty to be performed with great safety and comfort for the patient.

PROF. **DIEGO BASAN**

Dentist, specialist in Oral and Maxillofacial Surgery, Periodontics and Implantology.



TR1U-SD ∅0.5mm

· Selective supragingival scaling with diamond



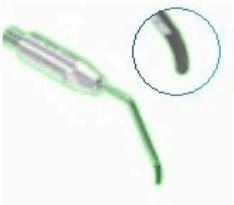
W1T-O 0.4mm x 2.5mm

· Conventional supragingival scaling



TR1-PK ∅0.5mm

· Clinical crown lengthening without flap opening (bone wear)



VR1-RPK ∅1.2mm

· Clinical crown lengthening without flap opening (bone wear)



VR1-LPK ∅1.2mm

· Clinical crown lengthening without flap opening (bone wear)



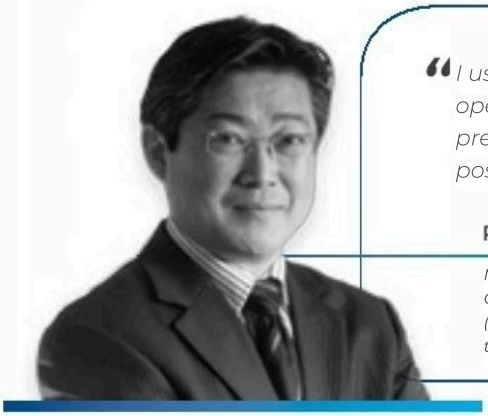
Photo - Prof. Paulo Kano: Detail of excess gingiva removed



Photo - Prof. Paulo Kano: Postoperative (1 day)



Photo - Prof. Marcelo de Faveri: Flapless surgery.



"I use CVDentus diamond inserts for crown lengthening without flap opening, without damage to the gingival tissue, with better precision, providing more comfort for the patient with a faster postoperative period.

PROF. **PAULO KANO**

Master's in Implantology, Director of the Paulo Kano Institute, Creator and Developer of the Clones method, Creator of the AST (Atomic Shell Technique) / SKIN Concept and dental prosthetics technician.



PROSTHETICS

Clinical PLUS



CAVITY

DentSurg PRO



C1

Ø 1.2mm

- Beveling
- Preparation finish line



C2T

Ø 1.6mm

- Post removal



CH1-D

1.0mm x 0.5mm

- Preparation finish line
- Preparation of veneers for contact lenses (narrow margin)



CH1-M

1.0mm x 0.5mm

- Preparation finish line
- Preparation of veneers for contact lenses (narrow margin)



CH4-D

1.6mm x 0.5mm

- Preparation finish line
- Preparation of veneers for contact lenses (wide margin)



CH4-M

1.6mm x 0.5mm

- Preparation finish line
- Preparation of veneers for contact lenses (wide margin)



CM4

Ø 3.7mm

- Cementation of contact lenses and prostheses



CR1

Ø 1.2mm

- Prosthetic preparation finish line (narrow margin)



CR4

Ø 1.6mm

- Prosthetic preparation finish line (wide margin)



CR4U

Ø 1.6mm

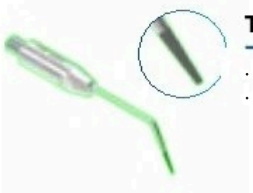
- Polishing of wide prosthetic margin
- Hybridization of immediate dentin sealing for veneers



CR1U

Ø 1.2mm

- Polishing of narrow prosthetic margin
- Hybridization of immediate dentin sealing for veneers



T1F

Ø 0.5mm

- Texturing
- CAD/CAM: Removal of temporary cement



Photo: Prof. Ricardo Bertholdi - Cementation



Photo: Prof. Rogério Marcondes - Cementation. Perfect seating of the ceramic restoration

“The CVDentus CM4 Cementation tip is a unique product in the world market, providing perfect seating of ceramic restorations, regardless of the viscosity of the cementing material.

PROF. **ROGÉRIO MARCONDES**

Specialist in Restorative Dentistry, Founding Member of Paraná Dental Team, Clinical Director of Dental Esthetic Center and Professor at Dental Esthetic Center Training.





TIF-E

Ø 0.5mm

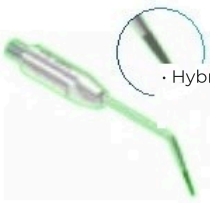
· Texturing



W1

0.5mm x 2.5mm

· Tooth sectioning
· Removal of veneers and contact lenses



TIU

Ø 0.5mm

· Alisamento e polimento de preparo
· Hybridization of immediate dentin sealing



W1-O

0.5mm x 2.5mm

· Tooth sectioning
· Removal of veneers and contact lenses



Photo - Prof. Renato Voss: Finish line.



Photo - Prof. Gustavo Greco: Veneer removal



“Using the piezo and the CVDentus W1 tip, I can completely remove veneers and contact lenses. I make a section and, with dilation, I perform the debonding. A very fast, simple procedure that does not damage the tooth.

PROF. GUSTAVO GRECO

Doctorate in Dentistry, Master's, Specialist in Prosthodontics, Coordinator of the Specialization course in Prosthodontics and Implantology - ABO-MG.



SURGERY

DentSurg PRO



SE1

Ø 3.9mm

· Sinus membrane elevation



SF3

0.28mm

· Osteogenic distraction
· Bone block removal (bone graft)
· Lateralization of the inferior alveolar nerve
· Alveolar ridge expansion
· Sectioning of ankylosed root
· Maxillary disjunction



SF38

0.28mm

· Osteogenic distraction
· Bone block removal (bone graft)
· Lateralization of the inferior alveolar nerve
· Alveolar ridge expansion (Split Crest)
· Sectioning of ankylosed root
· Maxillary disjunction



SF3F

0.40mm

· Maxillary sinus lift
· Alveolar ridge expansion



SF3F-O

0.40mm

· Autogenous bone harvesting
· Sectioning of ankylosed root
· Lateralization of the inferior alveolar nerve



SF3-L

0.40mm

· Bone block removal (bone graft)
· Maxillary sinus lift
· Lateralization of the inferior alveolar nerve



SF3-R

0.40mm

· Bone block removal (bone graft)
· Maxillary sinus lift
· Lateralization of the inferior alveolar nerve



SFS3-O

0.40mm

· Scraping and autogenous bone harvesting



"I use the SF3 and SF38 tips because of their high cutting power, always preserving the greatest amount of material with great predictability and safety.

PROF. **RICARDO SCOTTON**

Master's and Specialist in Implantology, Coordinator of the Advanced Surgery course In Surgery.



Photo - Prof. Ricardo Scotton: Lateralization of the Inferior Alveolar Nerve



Photo - Profs. Alan Costa and Shaban Burgoa: Bone-guided regularization



SFTV-O

0.40mm

- Periotomy in all teeth
- Delicate corticotomy



SLTV-O

0.40mm

- Flap elevation
- Periotomy of molars



SFTV2-O

0.40mm

- Opening of surgical window
- Osteotomy around apices or fractured implants



SCP

- Implant drilling
- Autogenous bone harvesting
- Trephine

Ø 3.0mm
3.5mm
3.8mm
4.1mm
4.8mm



SI

- Guide for SCP

Ø 3.0mm
3.5mm
3.8mm
4.1mm
4.8mm



SR2

Ø1.3mm

- Initial drilling for implant placement



SL1

0.28mm

- Osteotomy near soft tissue



SR4

Ø1.5mm

- Bone drilling

> *"The SFTV2-O tip can be used both for opening a surgical window, apicoectomy and osteotomy around apices and fractured implants, as well as for delicate osteotomies, allowing complicated situations to be resolved with minimal bone loss."*



PROF. TÚLIO VALCANAIA

Specialist, Master's and Doctorate in Oral and Maxillofacial Surgery and Traumatology, Partner and Director of IOA - Instituto de Odontologia das Américas and Partner and Director of IOA Miami - USA





Photos: Prof. Túlio Valcanaia - Detachment and extraction of the premolar

INFINITY

HIGH-SPEED ROTATION



FR4H

Ø1.6mm

· Prosthetic preparation, rounding of sharp angles for porcelain crowns



TR18H

Ø0.5mm

· Prosthetic preparation and beveling of the finish line



TR38H

Ø1.0mm

· Full crown preparation
· Prosthesis removal
· Removal of restorative material





Photos: Wear without damaging soft tissue

CLINICAL & SCIENTIFIC REFERENCES



Foto: Prof Ricardo Bertholdi

V. J. Trava Airoldi; E. J. Corat1; J. R. Moro.

STUDIES OF CVD DIAMOND APPLICATIONS WITH ULTRASOUND DEVICES IN DENTISTRY AND RELATED USES.

Brazilian Journal of Vacuum Applications, v. 25, n. 2, 71-74, 2006

Very precise, durable and quiet tools for many applications have been of great interest. Diamond-coated ultrasonic tools are excellent alternatives from this perspective, mainly for cutting hard materials. However, conventional technology for aggregating diamond powder with nickel metallic binders could not withstand ultrasonic energy. The diamond grains flake off easily and the tools are not durable enough while in contact with hard surfaces. This work presents studies on the deposition and coating of CVD diamond on different shapes of metallic substrates, with excellent adhesion, to obtain abrasion devices for an ultrasonic scaler used in dentistry. This type of technique is powerful for cutting and polishing hard materials with very good precision and less noise. Regarding dental cavity preparation, it presents additional advantages of less pain and no bleeding. The diamond coating was obtained using an Improved Hot-Filament Assisted Technique, with a high growth rate, using conventional mixtures of hydrogen and methane gas. Small special metal shanks machined in the shape of dental burs, with a pre-prepared surface, were coated with a thick CVD diamond film.

These tips were tested for their service life and cutting precision by abrasive wear of tooth enamel and glass surfaces using an ultrasonic scaler, showing very good performance. CVD ultrasonic diamond tips are about 30 times more durable than conventional ones. In all tests, the cutting regions show no damaged edges and the hole diameter maintains precise values as long as the tips last. Analyses of the morphology and roughness of the worn surface were evaluated by SEM and AFM techniques. An important application of this device in dentistry, presented for the first time, is also discussed.

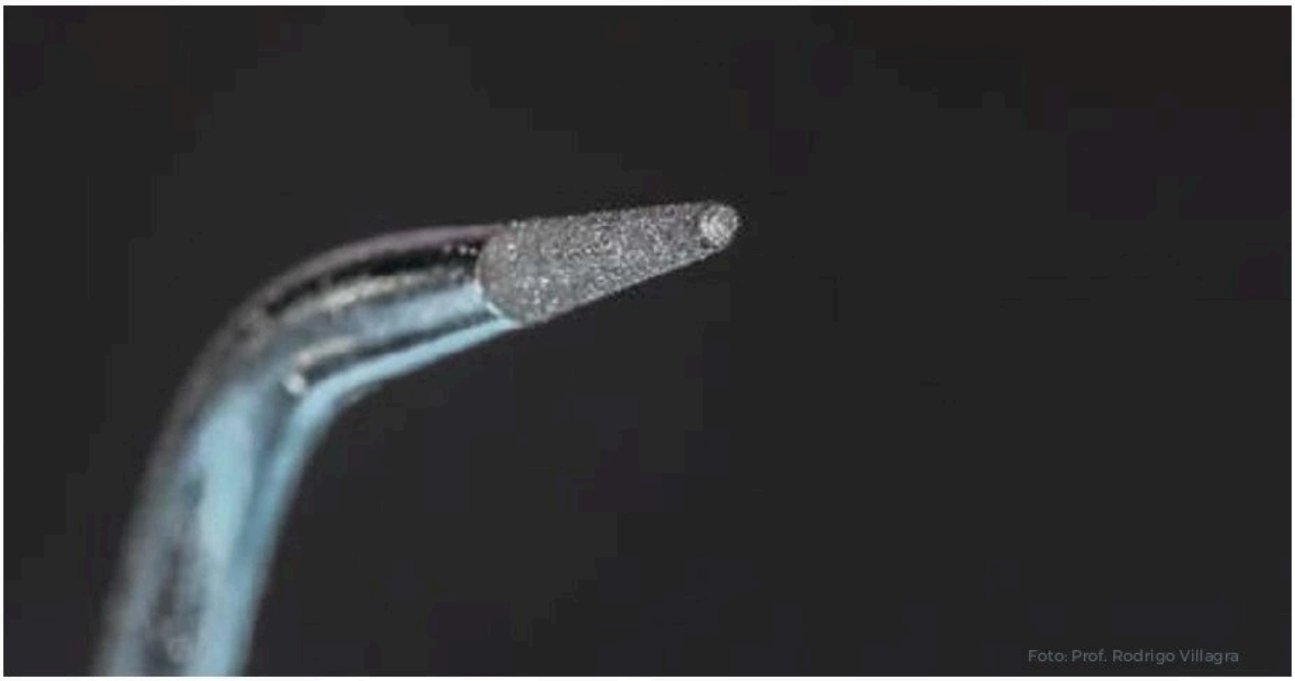


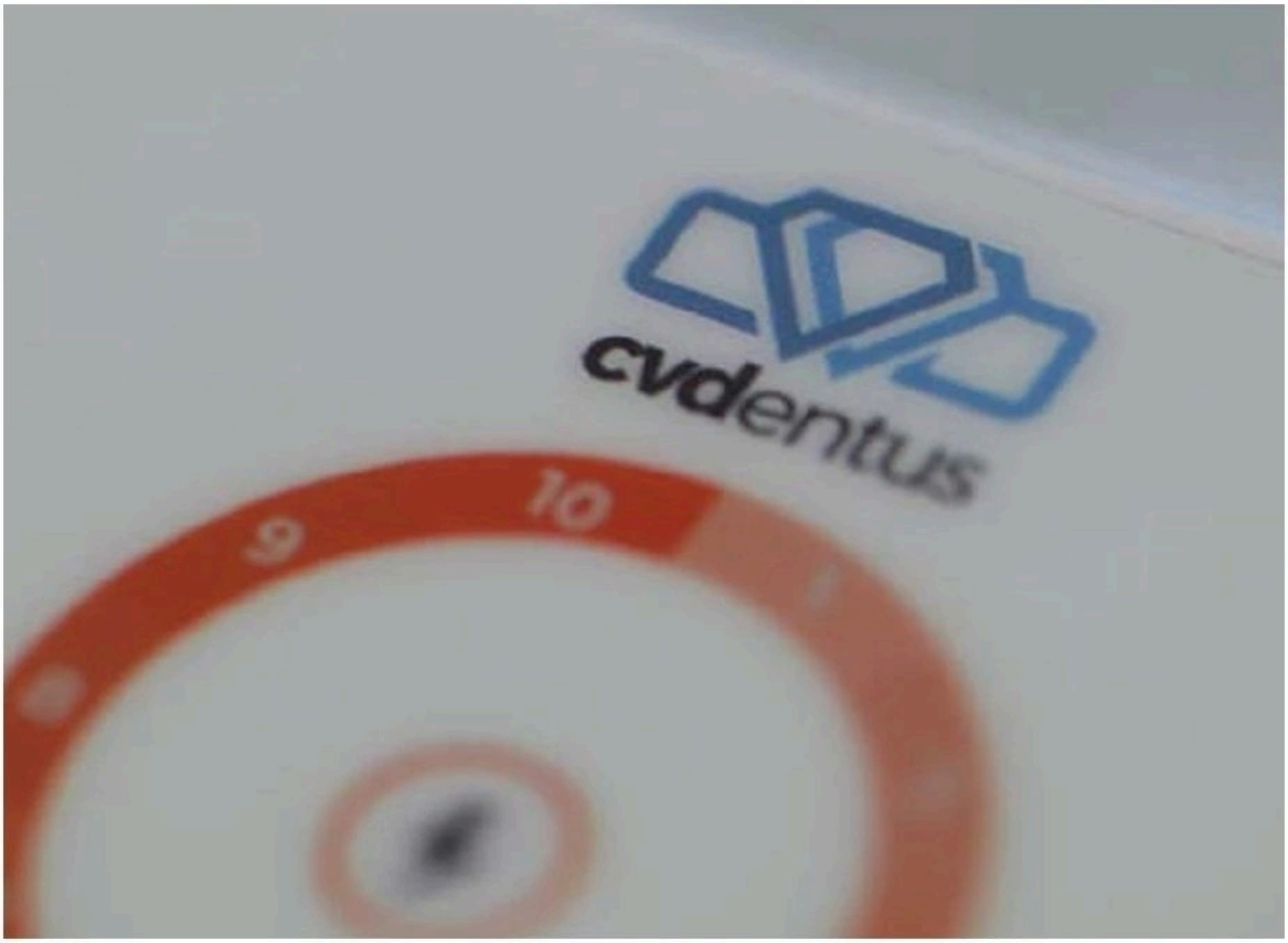
Foto: Prof. Rodrigo Villagra

Simone Di Salvo Mastrantonio; Juliana Oliveira Gondim; Érika Botelho Josgrilberg; Rita de Cássia Loiola Cordeiro.

REDUCING FEAR DURING DENTAL TREATMENT USING ULTRASONIC TIPS

Porto Alegre, v. 58, n. 1, p. 119-122, Jan./Mar. 2010

In recent years, new technologies have been developed for making cavity preparations, among which CVDentus® diamond tips (CVDVale, São José dos Campos, Brazil) stand out. These tips are produced by chemical vapor deposition, forming a single diamond stone, with greater durability than conventional diamond-coated tips. When coupled to the ultrasound device, they have several clinical applications in Dentistry with advantages over conventional rotary instruments, such as less pressure, noise, vibration and heating, in addition to reducing the need for local anesthesia, helping to reduce the patient's fear and anxiety. The objective of this work is to present a complete restorative dental treatment, performed with this system, in a pediatric patient with a previous history of uncooperative behavior. The use of this new technology offered greater comfort to the patient, enabling his or her reconditioning to dental treatment, besides having favored conservative cavity preparations. Index terms: anxiety about dental treatment; pediatric dentistry; ultrasound.



Marco Aurelio Paschoal; Rita de Cássia Loiola Cordeiro; Lourdes Santos-Pinto.

USE OF CVD ULTRASONIC TIPS IN THE MANAGEMENT OF THE PEDIATRIC PATIENT

J Health Sci Inst. 2011; 29(4):254-6

CVD® tips stand out for being used as an alternative to the rotary instrument for cavity preparation, since they provide safety and comfort to the patient. The objective of this work was to present two clinical cases of pediatric patients with caries lesions in the anterior teeth, in which the removal of carious tissue involved the use of CVD® tips coupled to ultrasound.

It was observed that this technology proved effective for readapting patients to the dental environment, in addition to being accessible to individuals with motor disorders, which prevent their travel to an appropriate clinical environment.



Michele Baffi Diniz; Renata de Medeiros Gianotto; Rita de Cássia Loiola Cordeiro.

REMOVAL OF CARIOUS TISSUE WITH ULTRASONIC CVD TIPS AS A CHILD MANAGEMENT STRATEGY

Revista Inst. Ciência Saúde 2008;26(2):263-6

Early childhood caries usually affects the upper incisors and primary molars in very young children, making treatment extremely difficult. CVD ultrasonic diamond tips can be used as an alternative to the conventional rotary instrument for making cavity preparations in situations such as the one described above, since they provide safety and comfort to the patient, in addition to being simple and easy to perform.

The objective of this work is to report the case of a 3-year-old patient who presented caries lesions on teeth 51, 52 and 62, whose treatment was the use of the CVD ultrasonic diamond tip to remove carious tissue followed by esthetic rehabilitation using the plastic matrix technique. It was observed that this new technology for making cavity preparations proved effective in safely preparing teeth affected by baby bottle caries, providing greater comfort and absence of painful symptoms in a very young patient.

Luciana Monti Lima; Cristiane Motisuki; Lourdes dos Santos-Pinto; Ary dos Santos-Pinto; Evaldo Jose Corat.

CUTTING CHARACTERISTICS OF DIAMOND-COATED DENTAL TIPS OBTAINED BY CVD TECHNOLOGY

Restorative Dentistry, 2006; 20(2):155-61

The objective of this study was to determine the cutting ability of diamond tips obtained by the chemical vapor deposition (CVD) process, associated with the ultrasound device in minimally invasive cavity preparation. A standardized cavity was prepared on the mesial and distal surfaces of 40 third molars, using cylindrical and spherical CVD diamond tips. Cutting ability was compared according to the type of substrate (enamel and dentin) and the direction of the movement performed with the tip.

The morphological characteristics, the width and depth of the cavities were analyzed and measured by scanning electron microscopy. Statistical analysis by the Kruskal-Wallis test ($p < 0.05$) revealed that the width and depth of the cavities were significantly greater in dentin. Wider cavities were obtained when the cylindrical CVD diamond tip was used, and deeper ones when the spherical tip was employed. The direction of the tip movement did not influence the size of the cavities, and the cuts produced by CVD diamond tips were precise and conservative



Luis Augusto Lupato Conrado; Egberto Munin; Evaldo José Corat; Vladimir Jesus TravaAiroldi.

NEW ULTRASONIC BUR COATED WITH CVD DIAMOND FOR CAVITY PREPARATION

Submitted to Journal of Operative Dentistry

This work reports a new concept of ultrasonic drilling of dental tissues. We present a new ultrasonic bur for dental applications, with a tip coated with chemical vapor deposited (CVD) diamond. The CVD process used during the manufacture of the bur ensures the durable nature of the diamond coating, bringing the existing idea of the ultrasonic dental bur to reality. CVD diamond is created on cylindrical molybdenum tips adapted for rotary and high-speed ultrasonic handpieces. Four healthy extracted molars were sectioned along their longitudinal axis to expose the dentinoenamel junction. These junctions were ground by conventional coarse and finishing diamond burs and CVD burs with a high-speed rotary system and a CVD ultrasonic bur. Scanning electron microscopy (SEM) was used to compare the surface finish of each technique. Twenty-five carious extracted molars with different types of caries were used to test cavity preparation in vitro with the CVD ultrasonic bur. We observed 52 clinical cavity preparations with the ultrasonic bur. The CVD ultrasonic bur efficiently cuts enamel and dentin with an excellent surface finish relatively free of a smear layer. The CVD bur is efficient and durable and eliminates the need for the abrasive paste reported in previously published work. Operator access is excellent and much better than with a high-speed rotary handpiece. This work indicates the CVD ultrasonic bur as an effective choice for painless and minimally invasive dental cavity preparation.

Carlos Augusto R. Carvalho; Ticiane C. Fagundes; Terezinha J. E. Barata; Vladimir Jesus Trava-Airoldi; Maria Fidela L. Navarro.

THE USE OF A CVD-COATED DIAMOND TIP COUPLED TO AN ULTRASONIC HANDPIECE IN TOOTH PREPARATION

Journal Compilation, Volume 19, Number 1, 2007



In recent decades, scientific developments in cutting instruments have changed the conventional techniques used to remove carious lesions. Ultrasound emerged as an alternative for caries removal in the 1950s. However, conventional technology for aggregating diamond powder with nickel metallic binders could not withstand ultrasonic energy. About 5 years ago, an alternative approach using chemical vapor deposition (CVD) resulted in synthetic diamond technology. CVD diamond burs are obtained with high adhesion of the diamond as a single stone on the metal surface with excellent abrasion performance. This technology allows the deposition of diamond with coalescent grain size on different substrate shapes. When connected to an ultrasonic handpiece, CVD diamond burs become an option for cavity preparation, maximizing the preservation of tooth structure. Potential advantages include reduced noise, minimal damage to gingival tissue, extended bur durability, improved access to the proximal cavity, reduced risk of reaching the adjacent tooth resulting from high inclination angles and minimal risk of metal contamination of the patient. These innovative instruments also potentially eliminate some problems related to the decreased cutting efficiency of conventional diamond burs. This clinical report presents the benefits of using CVD diamond burs together with an ultrasonic handpiece in the treatment of incipient caries. CVD diamond burs, coupled to an ultrasonic device, offer a promising alternative for the removal of carious lesions when ultraconservative cavity preparations are necessary. In addition, this system provides a less painful technique for caries removal, with minimal noise.

M. Youssef; A. Quinelato; F. Youssef; J. E. Pelizon Pelino; M. C. Salvadori; M. Mori.

CUTTING EFFICIENCY OF THE DENTAL SURFACE USING A HIGH-SPEED BUR, ULTRASOUND AND LASER

Laser Physics, 2008, Vol. 18, No. 4, pp. 472–477

We compared an ultrasound bur with a conventional one and a laser for cavity preparations. Human molars were embedded in resin and sliced for this study. Surface abrasion was performed by a high-speed instrument and ultrasound. Cavity preparation was initially performed with a high-speed diamond bur. After that, a 2.94 µm laser with 400 mJ/pulse at 4 Hz and a pulse width of 250 to 500 µs was applied to the tooth surface for 30 s in a sweeping motion.

The samples were analyzed by SEM. The surface abraded with the conventional bur showed removal of the structure with different grooves, presence of smear layer and occluded dentinal tubules. The surface worn with the CVD bur suggested a layer-by-layer removal process. The laser-irradiated surface showed a rough appearance with open tubules and absence of a smear layer. The results of this study suggest that a high-speed diamond bur, ultrasound and laser were able to perform the cavity preparation. However, the CVD bur showed higher surface quality.

Carlos Augusto R. Carvalho; Ticiane C. Fagundes; Terezinha J. E. Barata; Vladimir Jesus Trava-Airoldi; Maria Fidela L. Navarro.

THE USE OF CVD DIAMOND FOR ULTRACONSERVATIVE CAVITY PREPARATIONS: A REPORT OF TWO CASES

The present work aims to compare the roughness of enamel and dentin surfaces when prepared with high-speed diamond tips and when prepared with diamond burs for ultrasound. From the development of diamond tips by the CVD (Chemical Vapor Deposition) technique, it was possible to give greater resistance and durability to diamond tips, thus making it possible to adapt them to ultrasound devices, capable of promoting wear and cavity preparations in teeth. It becomes interesting to analyze the results obtained by these tips with those obtained with conventional diamond tips for high speed. Dentin and enamel samples were worn with conventional coarse-grit diamond burs, finishing burs, CVD tips for high speed and CVD tips for ultrasound, all samples being observed by scanning electron microscopy to evaluate the degree of roughness of their surfaces. It was observed that the samples prepared with CVD burs for ultrasound were those that produced the lowest degree of roughness.



Manon Eleonora de Ladalardo Martins; Mirian Rubio Faria; Maurício Rufaiel Matson.

MICROMORPHOLOGICAL ANALYSIS OF HUMAN DENTIN CONDITIONED BY SELF-ETCHING PRIMER VARYING THE PREPARATION TECHNIQUE: DIAMOND TIP, CARBIDE BUR AND CVDENTUS TIP

Pesq. Bras. Odontoped. Clin. Integr. João Pessoa, v. 6, n. 2, p. 161-166, mai/a

Objective: to evaluate, by means of SEM, the conditioning pattern of dentin after application of the self-etching primer of an adhesive system (Clearfil Liner Bond 2) and of 37% phosphoric acid, varying the preparation instrument. Method: 30 cavity preparations were performed in 5 recently extracted human molars, with 2 preparations with a diamond tip, 2 preparations with carbide burs at low speed and 2 preparations with an ultrasound-driven diamond tip (for each tooth). Each tooth received one type of treatment: no conditioning, conditioning with 37% phosphoric acid for 15 seconds or application of the self-etching primer. The specimens were analyzed by SEM, where images were taken at 200, 2000 and 5000 times magnification. Results: for the cases prepared with the CVDentus tip, we noticed a flake-like smear layer in addition to the presence of exposed dentinal tubules. In the preparations with conventional diamond tips and carbide bur, the smear layer was more compact, with presence of residues on the intertubular dentin. Conclusion: from this analysis we can state that with the use of CVDentus tips plus the application of a self-etching primer we have a less adherent smear layer, with a different dentin conditioning pattern than in conventional preparations.

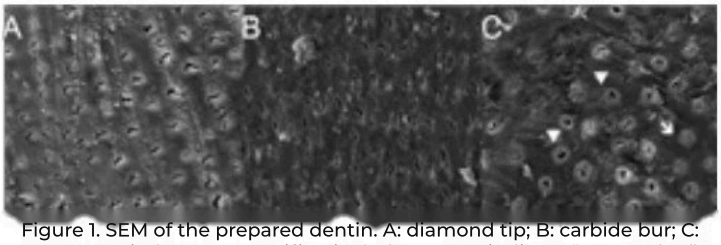


Figure 1. SEM of the prepared dentin. A: diamond tip; B: carbide bur; C: CVDentus tip (2000X magnification), the arrows indicate “smear plug”, the triangles indicate open tubules.

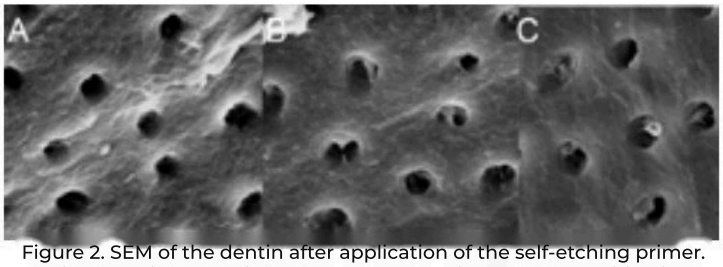


Figure 2. SEM of the dentin after application of the self-etching primer. A: diamond tip; B: carbide bur; C: CVDentus tip (5000X magnification).

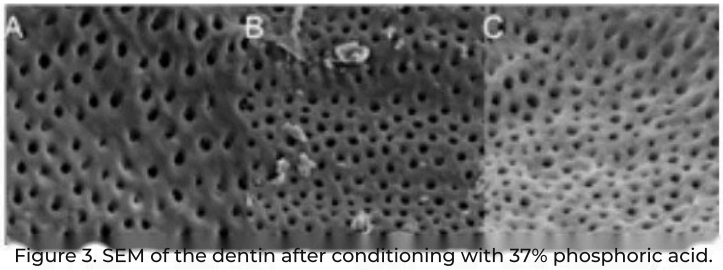


Figure 3. SEM of the dentin after conditioning with 37% phosphoric acid. A: diamond tip; B: carbide bur; C: CVDentus tip (2000X magnification).

Bruna Perdonsini Martini, Gabrielle de Souza Laynes, Aline Sebastiani.

USE OF THE PIEZOELECTRIC MOTOR IN MINOR ORAL SURGERY – CASE REPORT

Universidade Positivo, Curitiba, Paraná, Brazil – 2019.

Piezoelectricity consists of the ability of some crystals to generate electrical voltage in response to mechanical pressure. The surgical piezoelectric motor was developed by Horton in 1975 and improved by Vercelotti in 2007. Regarding minor oral surgeries, the use of the surgical piezoelectric motor has already been reported in cases of maxillary sinus lift, clinical crown lengthening, third molar extraction techniques, preparation for dental implant placement and removal of autogenous graft. The use of this device in surgeries presents several advantages such as no heating during osteotomy, excellent cutting properties and precision, preservation of soft tissues and reduced postoperative edema. The objective of this work is to report two cases of minor oral surgery using the DentSurg Pro device from CVDentus®, emphasizing the advantages of this equipment during the procedures and in the postoperative period.



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